How-To Library

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Performance Tuning and Sizing Guidelines for Microsoft Azure Synapse SQL Connector

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Abstract

You can tune the hardware parameters, database parameters, application server parameters, and Informatica mapping parameters to optimize the performance of Microsoft Azure Synapse SQL Connector. This article describes general reference guidelines, best practices, and case studies to help you tune the performance of Microsoft Azure Synapse SQL Connector.

Supported Versions

- Cloud Data Integration Microsoft Azure Synapse SQL Connector

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Overview

Performance tuning is an iterative process in which you analyze the performance, use guidelines to estimate and define parameters that impact the performance, and monitor and adjust the results as required.

This document describes the key hardware, batch size, and mapping parameters that you can tune and the various pushdown scenarios that you can configure to optimize the performance of Microsoft Azure Synapse SQL Connector.

When you apply pushdown optimization, the task pushes transformation logic to the source or target database based on the optimization type you specify in the task properties. Data Integration translates the transformation logic into a PolyBase query or SQL query. The task sends the query to the database and the database executes the query.

Note: The performance testing results listed in this article are based on observations in an internal Informatica environment using data from real-world scenarios. The performance of Microsoft Azure Synapse SQL Connector might vary based on individual environments and other parameters even when you use the same data.

Pushdown optimization types

You can configure the following types of pushdown optimization to optimize your Microsoft Azure Synapse SQL mapping:

Source-side pushdown optimization

When you run a task enabled with source pushdown optimization, the task pushes down as much as the transformation logic as possible to process in the source database. Source pushdown optimization does not include the Target transformations.

Full pushdown optimization

When you run a task enabled with full pushdown optimization, the task pushes as much of the transformation logic as possible to process in the target database.

Partial pushdown optimization

Partial pushdown uses a combination of source and full pushdown. When you run a mapping with full pushdown optimization, the mapping attempts to perform full pushdown in the beginning and pushes the source, target, and other transformations configured in the mapping to the target.

If all the transformations in the mapping are supported for pushdown optimization, the task pushes down the entire mapping logic to the target database. If there is an unsupported transformation or unsupported function in the mapping while attempting the pushdown, full pushdown is converted into source pushdown. In this case, the task employs partial pushdown optimization and pushes down the mapping logic to the source database until the point where the transformation is supported for pushdown optimization in the mapping.

Performance sizing parameters

You can tune the hardware parameters based on the data set and use the recommended hardware capacities to optimize the performance of the machine where the Secure Agent runs.

Performance sizing depends on the size of the data set. Consider the following sizing recommendations for the different types of data sets:

Small data set

A data set is considered to be a small data set if you read from and write to less than or equal to 750 MB of data.

For a small data set, you can read 750 MB of data from a flat file source and write to Microsoft Azure Synapse SQL in `00:02:12` minutes. Read data from Microsoft Azure Synapse SQL and write data to a flat file target in `00:02:48` minutes.

The following table describes the sizing recommendations for the system that processes small data sets:

Requirements	Infrastructure
Services	Informatica Intelligent Cloud Services Agent
VCPUs	1
Memory	4 GB
Storage	50GB

Medium data set

A data set is considered to be a medium data set if you read from and write to less than or equal to 7.5 GB of data.

For a medium data set, you can read 7.5 GB of data from a flat file source and write to Microsoft Azure Synapse SQL in `00:15:45` minutes. Read data from Microsoft Azure Synapse SQL and write data to a flat file target in `00:18:07` minutes.

The following table describes the sizing recommendations for the system that processes medium data sets:

Requirements	Infrastructure
Services	Informatica Intelligent Cloud Services Agent
VCPUs	2
Memory	8 GB
Storage	100 GB

Large data set

A data set is considered to be a large data set if you read from and write to less than or equal to 75 GB of data.

For a large data set, you can read 75 GB of data from a flat file source and write to Microsoft Azure Synapse SQL in `02:37:58` hours. Read data from Microsoft Azure Synapse SQL and write data to a flat file target in `03:13:54` hours.

The following table describes the sizing recommendations for the system that processes large data sets:

Requirements	Infrastructure
Services	Informatica Intelligent Cloud Services Agent
VCPUs	8
Memory	28 GB
Storage	250 GB

Performance tuning parameters

You can optimize the performance of Microsoft Azure Synapse SQL mappings by tuning the following areas:

- Tune the hardware
- Tune the mapping
  - Tune the reader performance
  - Tune the writer performance
  - Local flat file staging
  - Disk input and output

Tune the hardware

You can tune the following hardware parameters to optimize the performance of the machine where the Secure Agent runs:

- CPU frequency

- NIC card ring buffer size

CPU frequency

Dynamic frequency scaling adjusts the frequency of the processor on-the-fly either for power savings or to reduce heat. Ensure that the CPU operates at least at the base frequency.

When CPUs are underclocked, where they run below the base frequency, the performance degrades by 30% to 40%. Informatica recommends that you work with your IT system administrator to ensure that all the nodes on the cluster are configured to run at their supported base frequency.

To tune the CPU frequency for Intel multicore processors, perform the following steps:

1. Run the `lscpu` command to determine the current CPU frequency, base CPU frequency, and the maximum CPU frequency that the processor supports.
2. Request your system administrator to perform the following tasks:
   - Increase the CPU frequency to the supported base frequency.
   - Change the power management setting to **OS Control** at the BIOS level.
3. Run CPU-intensive tests to monitor the CPU frequency in real time and adjust the frequency for improved performance. On Red Hat operating systems, you can install a monitoring tool such as cpupower.
4. Work with your IT department to ensure that the CPU frequency and power management settings are persisted even for future system reboots.

NIC card ring buffer size

NIC configuration is a key factor in network performance tuning. When you deal with large volumes of data, it is crucial that you tune the Receive (RX) and Transmit (TX) ring buffer size. The ring buffers contain descriptors or pointers to the socket kernel buffers that hold the packet data.

You can run the `ethtool` command to determine the current configuration.

For example, run the following command:

```
# ethtool -g eth0
```

The following sections show a sample output:

```
Ring parameters for eth0:

Pre-set maximums:

RX: 2040

RX Mini: 0

RX Jumbo: 8160

TX: 255

Current hardware settings:

RX: 255

RX Mini: 0

RX Jumbo: 0

TX: 255
```

The **Pre-set maximums** section shows the maximum values that you can set for each parameter. The **Current hardware settings** section shows the current configuration details.

A large buffer size leads to low latency. However, low latency comes at the cost of throughput. For greater throughput, you must configure large buffer ring sizes for RX and TX.

Informatica recommends that you use the `ethtool` command to determine the current hardware settings and the maximum supported values. Then, set the values based on the maximum values that are supported for each operating system.

For example, if the maximum supported value for RX is 2040, you can use the following `ethtool` command to set the RX value to 2040:

```
# ethtool -G eth0 RX 2040
```

If you set a low ring buffer size for data transfer, packets might get dropped. To find out if packets were dropped, you can use the `netstat` and `ifconfig` commands.

The following image shows a sample output of the `netstat` command:

```
$ netstat -ip
Kernel Interface table
Iface       MTU Met    RX-OK RX-ERR RX-DRP RX-OVR    TX-OK TX-ERR TX-DRP TX-OVR Flg
bond0      1500   0 15923289077      0  42154      0 26491595457      0      0      0 BMmRU
eth0       1500   0 10583980195      0  39877      0 12388943310      0      0      0 BMsRU
eth1       1500   0 5339308882       0   2277      0 14102652147      0      0      0 BMsRU
eth2       1500   0 17097891         0      0      0 27133559         0      0      0 BMRU
lo        16436   0 3678110101       0      0      0 3678110101       0      0      0 LRU
```

The RX-DRP column indicates the number of packets that were dropped. Set the RX value such that no packets get dropped and the RX-DRP column shows the values as 0. You might need to test several values to optimize the performance.

The following image shows a sample output of the `ifconfig` command:

```
$ /sbin/ifconfig
bond0     Link encap:Ethernet  HWaddr 78:2B:CB:4F:62:9F
          inet addr:10.1.41.215  Bcast:10.1.43.255  Mask:255.255.252.0
          inet6 addr: fe80::7a2b:cbff:fe4f:629f/64 Scope:Link
          UP BROADCAST RUNNING MASTER MULTICAST  MTU:1500  Metric:1
          RX packets:15923290357 errors:0 dropped:42154 overruns:0 frame:0
          TX packets:26491595749 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:17166401902541 (15.6 TiB)  TX bytes:34239226692992 (31.1 TiB)

eth0      Link encap:Ethernet  HWaddr 78:2B:CB:4F:62:9F
          UP BROADCAST RUNNING SLAVE MULTICAST  MTU:1500  Metric:1
          RX packets:10583980886 errors:0 dropped:39877 overruns:0 frame:0
          TX packets:12388943313 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:11210108051986 (10.1 TiB)  TX bytes:16138022061439 (14.6 TiB)
          Interrupt:106 Memory:d8000000-d8012800

eth1      Link encap:Ethernet  HWaddr 78:2B:CB:4F:62:9F
          UP BROADCAST RUNNING SLAVE MULTICAST  MTU:1500  Metric:1
          RX packets:5339309471 errors:0 dropped:2277 overruns:0 frame:0
          TX packets:14102652436 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5956293850555 (5.4 TiB)  TX bytes:18101204631553 (16.4 TiB)
          Interrupt:114 Memory:da000000-da012800
```

The status messages indicate the number of packets that were dropped.

Tune the mapping

You can optimize the performance of Microsoft Azure Synapse SQL Connector by tuning the reader and writer mapping.

Tune the reader performance

The performance of a read operation depends on the heap size of the Secure Agent and the advanced source properties.

Tune the heap size for the Secure Agent

The default heap size is 64 MB. To increase performance and avoid runtime environment memory issues, configure the Java heap size for the Informatica Cloud Secure Agent.

Perform the following steps to increase the heap size:

1. Log in to Informatica Intelligent Cloud Services.
2. Select **Administrator** > **Runtime Environments**.
3. On the **Runtime Environments** page, select the Secure Agent for which you want to increase memory from the list of available Secure Agents.
4. On the upper-right corner of the page, click **Edit**.
5. In the **System Configuration Details** section, select the **Type** as **DTM** for the Data Integration Server.
6. Enter **JVMOption1** as **-Xmx1024m**.
7. Click **Save**.

Tune the advanced source properties

You can optimize the performance by configuring the following advanced source properties:

Block size

The size of the blocks of data used to divide a large file or object into smaller parts of the specified block size to process data in parallel. The default block size is 8388608 bytes or 8 MB. The block size must be less than or equal to the calculated optimal block size.

The following formula is used to calculate the optimal block size for the read operation:

```
Calculated_Optimized_Block_Size =(Maximum_heap_allocated * 0.2)/(Concurrent_Threads*2)
```

If the block size for download is greater than the calculated optimal block size, the calculated optimal block size is used as the block size to download data from a Microsoft Azure Synapse SQL source to the Secure Agent hosted virtual machine.

Concurrent threads

The number of concurrent connections to extract data from the Microsoft Azure Synapse SQL. The concurrent threads must be less than or equal to twice the number of VCPUs on the Secure Agent system.

Timeout interval

The number of seconds to wait when attempting to connect to the server. The default timeout interval is 60 seconds and the maximum value is 3600 seconds.

Note: If the timeout interval is set to default, the mapping task might fail with the following error: `error: java.util.concurrent.ExecutionException: java.lang.RuntimeException: java.net.SocketException: Connection reset`

To resolve this issue, set the following values for the timeout interval based on the size of the data set:

Parameter	Small data set	Medium data set	Large data set
Timeout interval	300	900	1800

Case study: Impact of reader tuning on download time

This case study describes the impact of reader tuning on the download time.

The following configurations are considered for the case study:

- The source and target are in the same region.
- The Secure Agent is hosted on an Azure Virtual Machine and is in the same region as the source and target.
- There are no transformations between the source and target.
- The number of Data Warehouse Units in Microsoft Azure Synapse SQL is set to 400c.

Environment

Virtual Machine	Azure Virtual Machine
VCPU	8 VCPUs
Memory	28 GB RAM

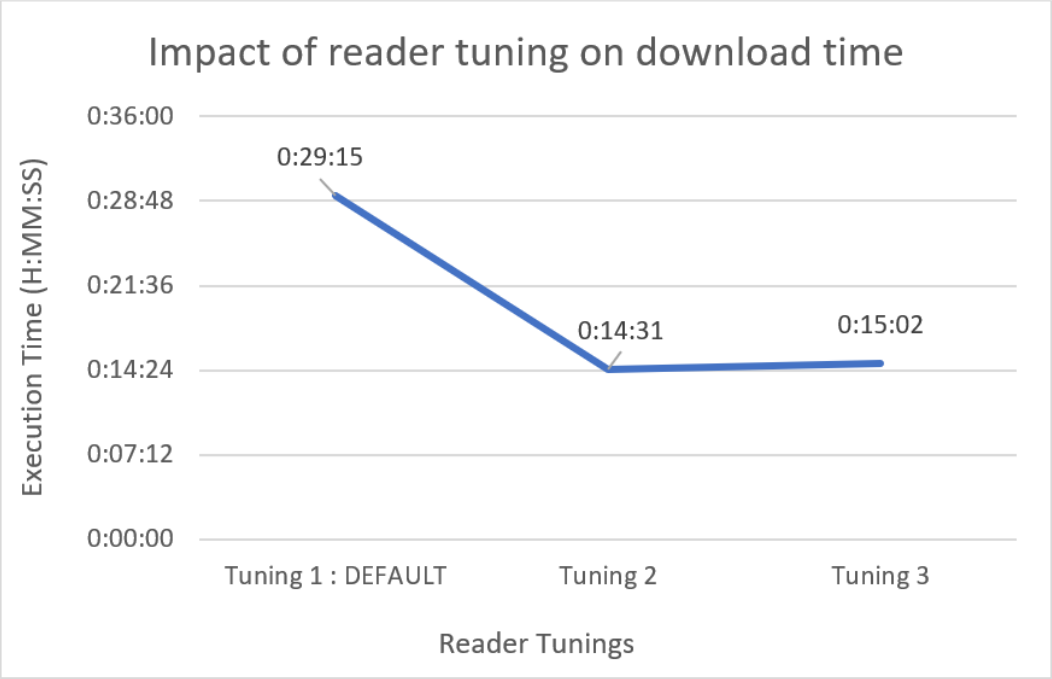
Performance table

The following table shows the different sets of reader tuning for a 75 GB data set with 600 million records:

Reader tuning	Given heap (MB)/ Maximum heap allocated (MB)	Block size (MB/Bytes)	Concurrent threads	Timeout interval
Tuning 1 (default)	64(default)/59	8/8388608	4	1800
Tuning 2	1024/910	8/8388608	10	1800
Tuning 3	2048/1820	16/16777216	10	1800

Performance graph

The following graph shows the impact of reader tuning on the download time for a 75 GB data set with 600 million records:



Conclusion

Based on the case study, it is observed that when you tune the reader, the performance improves on the second tuning.

The following table shows the tuning recommendations for the data sets:

Small data set	Medium data set	Large data set	Data set larger than 75 GB
Tuning 1 (default)	Tuning 2	Tuning 2	Tuning 3

Note: The download time depends on the network bandwidth and the distance between the geographical location of the source and target. This tuning is applicable only to the reader performance.

Tune the writer performance

You can tune the following Microsoft Azure Synapse SQL parameters to improve the writer performance:

Gzip compression

To move large datasets across the network, you can compress the file before uploading it to the network to minimize the upload time and network utilization. File compression uses ~90% of the CPU, as it spawns multiple threads to compress a file. The CPU utilization increases for a very short duration of time, as compression is 8 to 10 times faster with the parallel gzip implementation.

You can enable the gzip compression in the advanced target properties for a mass ingestion task.

Note: The time to upload a file depends on the network bandwidth of Secure Agent hosted virtual machine and the distance between the geographical location of the Secure Agent hosted virtual machine and the staging location.

Batch size tuning

When you have large data sets, set the batch size equal to or greater than the number of rows in the source for better performance. Batch size tuning improves the performance by 40% as compared to the default batch size for large data sets (600m records).

The following table shows the recommended batch size for the data sets:

Parameter	Small data set	Medium data set	Large data set
Batch size	Records <= 2 million: Default Records > 2million: 60 million	60 million	100 million

Local flat file staging

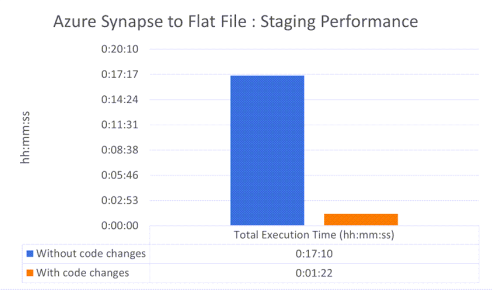
Informatica has redesigned the way Data Integration stages the data when you read from and write to an Microsoft Azure Synapse SQL target. This has significantly improved the performance in the local staging execution time.

You can optimize the mapping performance by setting the staging property in the Secure Agent to create a flat file in a temporary folder to stage the data locally before reading from or writing to Microsoft Azure Synapse SQL.

Reading from a Microsoft Azure Synapse SQL source

When you set the staging property *INFA_DTM_RDR_STAGING_ENABLED_CONNECTORS* for the Tomcat to the plugin ID of the Microsoft Azure Synapse SQL Connector in the Secure Agent properties, Data Integration, by default, creates a flat file locally to stage the data from the Microsoft Azure Synapse SQL source and then reads the data from the staging file. You can find the plugin ID in the manifest file located in the following directory: `<Secure Agent installation directory>/downloads/<Microsoft Azure Synapse SQL package>/CCIManifest`. For more information on how to set the property, see the help for Microsoft Azure Synapse SQL Connector.

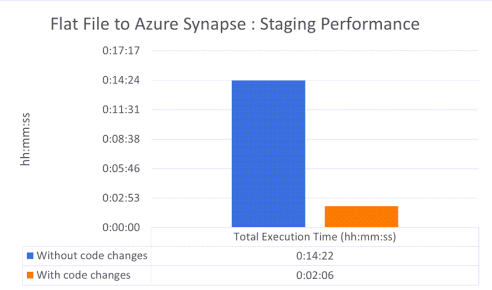
The following graph represents the improvements made to Microsoft Azure Synapse SQL local staging execution time for Microsoft Azure Synapse SQL to flat file passthrough mapping:



Writing to a Microsoft Azure Synapse SQL target

When you set the staging property *INFA_DTM_STAGING_ENABLED_CONNECTORS* for the Tomcat to the plugin ID of Microsoft Azure Synapse SQL Connector in the Secure Agent properties, Data Integration, by default, creates a flat file locally to stage the data and then loads the data from the staging file to the Microsoft Azure Synapse SQL target. You can find the plugin ID in the manifest file located in the following directory: `<Secure Agent installation directory>/downloads/<Microsoft Azure Synapse SQL package>/CCIManifest`. For more information on how to set the property, see the help for Microsoft Azure Synapse SQL Connector.

The following graph represents the improvements made to Microsoft Azure Synapse SQL local staging execution time for a flat file to Microsoft Azure Synapse SQL passthrough mapping:



Informatica recommends that you add three processor cores to the agent machine to increase the processing power of the mapping.

The following table shows the resource sizing in the old and new environment for a mapping that reads from a flat file source and writes to a Microsoft Azure Synapse SQL target:

Staging	CPU requirement (cores)	Memory requirement (GB)
Without code changes	3	2 GB (JVMHeap - 1 GB and DTM Buffer Size 1 GB)
With code changes	3	2 GB (JVMHeap - 1 GB and DTM Buffer Size 1 GB)

When you use three cores, the performance is optimized and the local staging execution time is far lesser. Though Informatica recommends that you include three cores for a Microsoft Azure Synapse SQL mapping execution, it is not mandatory. However, when you include lesser cores, the mapping performance is impacted but it does not result in execution failures.

Disk input and output

Azure virtual machines have Input/output Operations Per Second (IOPS) and throughput performance limits based on the virtual machine type and size.

OS disks and data disks can be attached to virtual machines. The data disks have their own IOPS and throughput limit. IOPS is the number of requests that the application sends to the storage disks in one second. An input/output operation could be read or write, sequential, or random. We must design the application in order to use the most optimal IOPS.

When you attach a premium storage disk to a virtual machine, Azure provisions for a fixed number of IOPS as per the disk specification. For example, a P50 disk provisions 7500 IOPS.

Throughput or bandwidth is the amount of data that the application sends to the storage disks in a specified interval. If the application performs large number of input and output operations, it needs high throughput.

Hence, it is important to determine the optimal throughput and IOPS values that the application requires.

The following case study shows the impact of disk input and output on the performance:

Case study

In the case study, the source is Microsoft Azure Synapse SQL and the target is a flat file. Four partitions are configured in the Microsoft Azure Synapse SQL source table that would cause sustained input and output load on the secure agent machine during local staging.

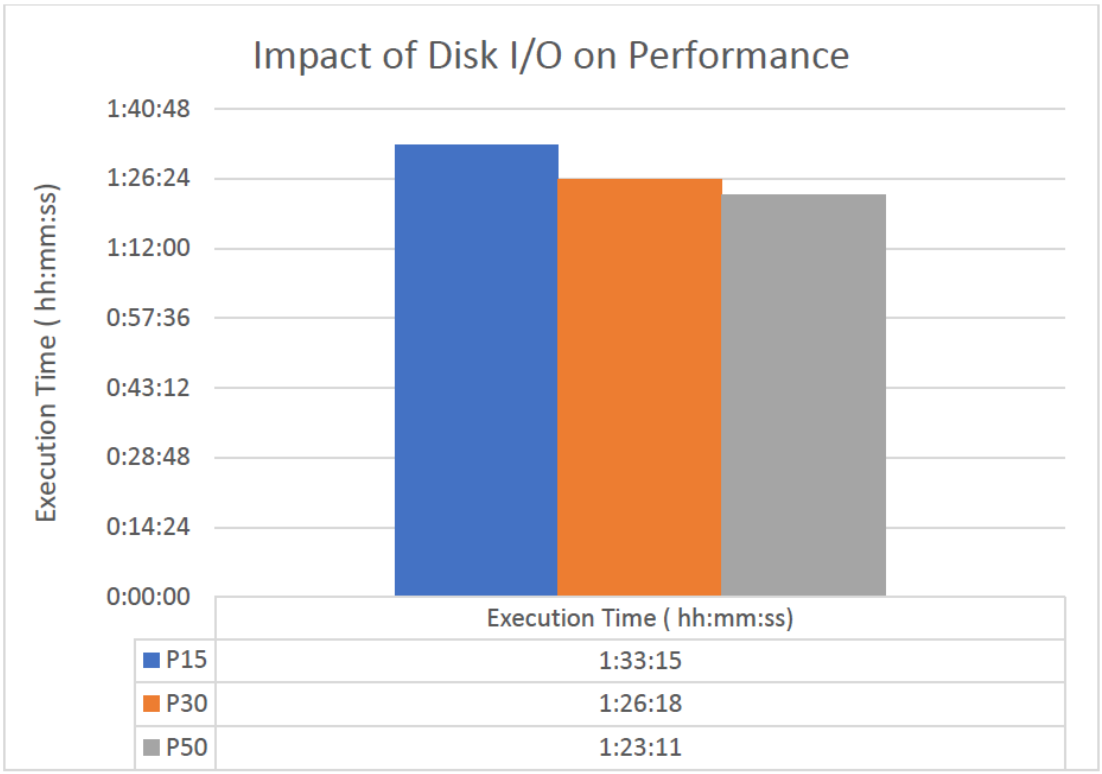
A custom data disk of 256 GB is used based on the data volume to be processed.

When you create a disk, the performance of the data disk is set in the form of its performance tier. The performance tier determines the IOPS and throughput limit. To increase the throughput, you can modify the performance tier of the data disk to different levels.

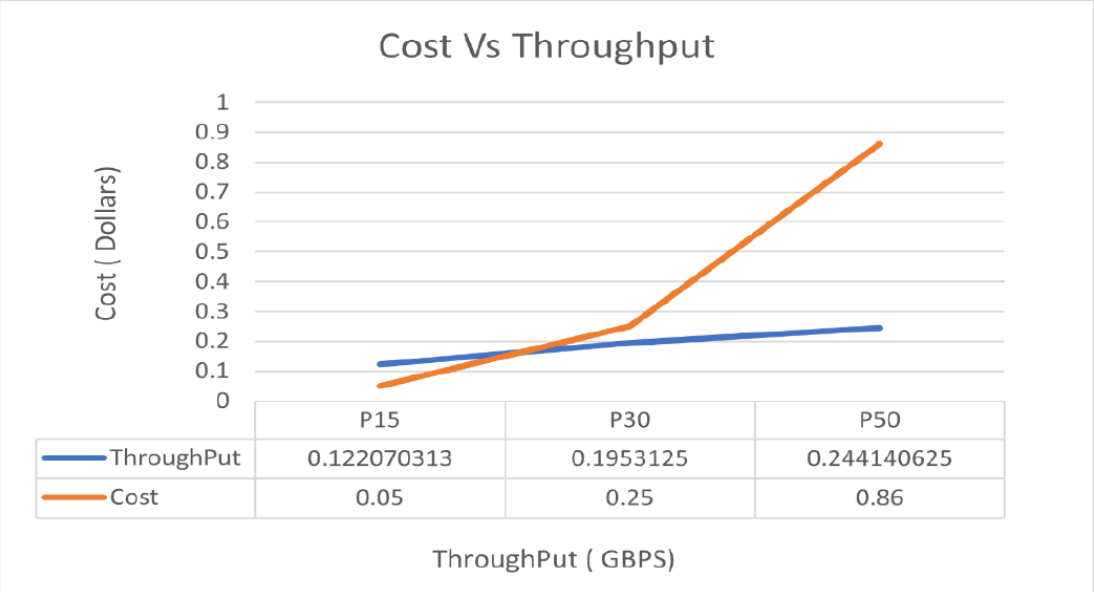
In the case study, the performance tier of the 256 GB data disk is modified across three levels P15, P30, and P50 that provide the following provisioned IOPS:

Performance tier	Provisioned IOPS (in Mbps)
P15	1100
P30	5000
P50	7500

The following image shows the impact of disk input and output on the performance:



The following image shows the impact of throughput on the cost:



Conclusion

- Based on the case study, it is observed that disk throughput plays a significant role in improving the performance.
  The staging execution time and the target execution time is reduced when the disk input and output is tuned. The disk with a throughput of 250 Mbps (P50) shows optimal performance. Further tuning does not yield much performance benefit.

  When you execute the same scenario with 16 partitions, the disk saturation reaches ~50%.
- A two fold increase in throughput from P15 to P30 results in an increase in the cost by 17x.
  The cost incurred includes only the disk cost and does not include the compute costs involved. The cost is approximated based on the total monthly pricing for the chosen disks as per the Azure disk pricing.

Summary

- To achieve optimal performance, it is critical to choose the disk size and the performance tier associated with it based on the application needs to achieve the most optimal disk throughput.
- Increasing the performance tier associated with the disk does not always guarantee an increase in the throughput, but may instead incur heavy costs.
- If the application throughput is critical, it is advisable to opt for larger disk sizes and higher performance tiers at an additional disk cost to improve the overall performance.

Best practices for performance tuning

When you tune the reader or writer, you can employ the following best practices to improve the performance:

- Host the Secure Agent on an Azure Virtual Machine. Azure provides the flexibility to scale or descale based on the workload and provides the VM resource monitoring.
- For data migration across networks, deploy the Secure Agent close to the source object.
- For data migration within a network, deploy the Secure Agent in the same region or network as the source and target object.
- If no transformations are required to process the data, use mass ingestion tasks to transfer a large number of files.

- Microsoft Azure Synapse SQL Gen2 provides better performance than Microsoft Azure Synapse SQL Gen1.

You can tune the following components of Microsoft Azure Virtual Machine that hosts the Secure Agent to improve the reader and writer performance:

Microsoft Azure Synapse SQL Gen2 cDWU tuning

Consider the following tuning recommendations for Microsoft Azure Synapse SQL:

Parameter	Small data set	Medium data set	Large data set
Microsoft Azure Synapse SQL units	DW100	DW350 or above	DW1000 or above

You can scale the Microsoft Azure Synapse SQL instance as per Microsoft Azure recommendations. For more information, see https://docs.microsoft.com/en-us/azure/synapse-analytics/sql-data-warehouse/what-is-a-data-warehouse-unit-dwu-cdwu.

Note: You must not perform scale-up or scale-down operations on Microsoft Azure Synapse SQL when a mapping is in progress, else the system kills all open sessions or queries and rolls back any open transactions to ensure a consistent state. Ensure that there are no queries running in Microsoft Azure Synapse SQL before you scale Microsoft Azure Synapse SQL.

Concurrent connections to Blob Store

When you read data from or write data to Microsoft Azure Synapse SQL, you can specify the number of concurrent connections to Blob Store.

- In a read operation, you can specify the number of concurrent connections to Blob Store to download files from Microsoft Azure Data Lake Storage Gen2 or Microsoft Azure Blob Storage.
- In a write operation, you can specify the number of concurrent connections to the Blob Store to upload files to Microsoft Azure Data Lake Storage Gen2 or Microsoft Azure Blob Storage.

The default number of concurrent connections set for a Microsoft Azure Synapse SQL object is 4 and the maximum value is 10.

You can improve the performance of reading or writing data to Microsoft Azure Synapse SQL by increasing the number of concurrent connections to Blob Store.

The following table shows the recommended number of concurrent connections for different data sets based on the staging location:

Parameters	Small data set	Medium data set	Large data set
Number of concurrent connections to Blob Store when the files are staged in Microsoft Azure Data Lake Storage Gen2	4	10	10
Number of concurrent connections to Blob Store when the files are staged in Microsoft Azure Blob Storage	4	10	10

Note: Ensure the following condition for the Secure Agent hosted virtual machine:

Number of CPU cores >= Number of concurrent connections to Blob Store

Microsoft Azure Virtual Machine

You can use the following components of a Microsoft Azure Virtual Machine that hosts the Secure Agent:

- **Virtual machine type** - Select the virtual machine type based on the workload and kind of operations to be run on the data set.
  For more information about virtual machine sizes, see https://docs.microsoft.com/en-us/azure/virtual-machines/windows/sizes?toc=/azure/virtual-machines/windows/toc.json
- **Disks** - The data is staged in the Secure Agent hosted virtual machine. Ensure that the size of the OS disk is greater than the size of data set. For better performance, use the disks with high IOPS and throughput.
  For more information about types of disks, see https://docs.microsoft.com/en-us/azure/virtual-machines/windows/disks-types

  The following table shows the recommended disk type for different data sets:

Parameter	Small data set	Medium data set	Large data set
Disk type	Standard HDD >= S10	Premium SSD >= P4	Premium SSD >= P10

Staging files

You can use Microsoft Azure Blob Storage or Microsoft Azure Data Lake Storage Gen2 as temporary storage for staging files.

Consider the following recommendations for staging files:

- For a Microsoft Azure Data Lake Storage Gen2 source or target, use Microsoft Azure Data Lake Storage Gen2 to stage files. Similarly, for a Microsoft Azure Blob Storage source or target, use Microsoft Azure Blob Storage to stage files.
- Informatica recommends that you have the source and staging location in the same geographical location or region.

Scalability test

Use the scalability test to understand the effects of increasing the data volume on the execution time by keeping the mapping level configurations, virtual machine size, and DWU constant.

Performance is linearly scalable when increase in the data volume is linear to the increase in the execution time.

Reader scalability test

The reader scalability test describes the effect of increasing the data volume on the execution time of a mapping when you read data from Microsoft Azure Synapse SQL.

The following configurations are considered for the test:

- The source is Microsoft Azure Synapse SQL hosted in the US-EAST-2 region. The target is a flat file data object on an Azure Linux Virtual Machine hosted in the US-EAST-2 region.
- All mappings are passthrough. There are no transformations between the source and target.
- The data set size for small, medium, and large data sets is defined based on the sizing recommendations.
- The Secure Agent is hosted on an Azure Virtual Machine in US-EAST-2 region for all the data sets.
- The files are staged in Microsoft Azure Blob Storage.
- The number of Data Warehouse Units in Microsoft Azure Synapse SQL is set to 400c.

- The following table describes the parameter values considered for the test:

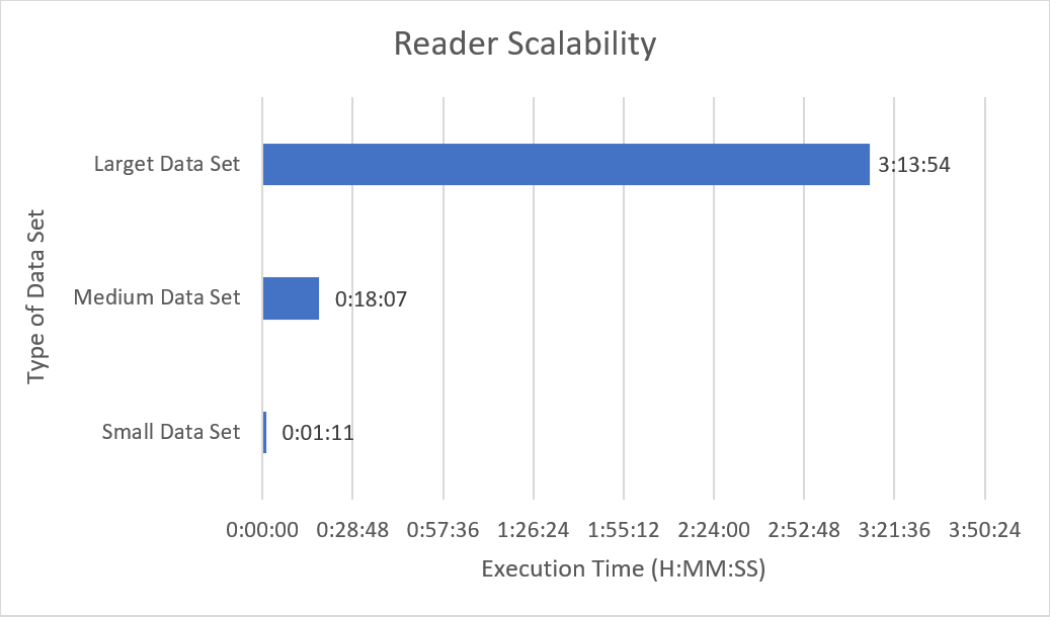
Tuning	Given heap (MB) / Maximum heap allocated(MB)	Block size (MB/Bytes)	Concurrent threads
Tuning 2	1024/910	8/8388608	10

Environment

Virtual Machine	Azure Virtual Machine
VCPU	8 VCPUs
Memory	28 GB RAM

Performance graph

The following graph shows the effect of increasing the data volume on the execution time in a reader mapping:



Conclusion

Based on the case study, linear scalability is observed for the reader mapping. When you increase the data volume, the execution time also increases linearly.

Writer scalability test

The writer scalability test describes the effect of increasing the data volume on the execution time of a mapping when you write data to Microsoft Azure Synapse SQL.

The following configurations are considered for the test:

- The source is a flat file data object on an Azure Linux Virtual Machine hosted in the US-EAST-2 region. The target is Microsoft Azure Synapse SQL hosted in the US-EAST-2 region.
- All mappings are passthrough. There are no transformations between the source and target.

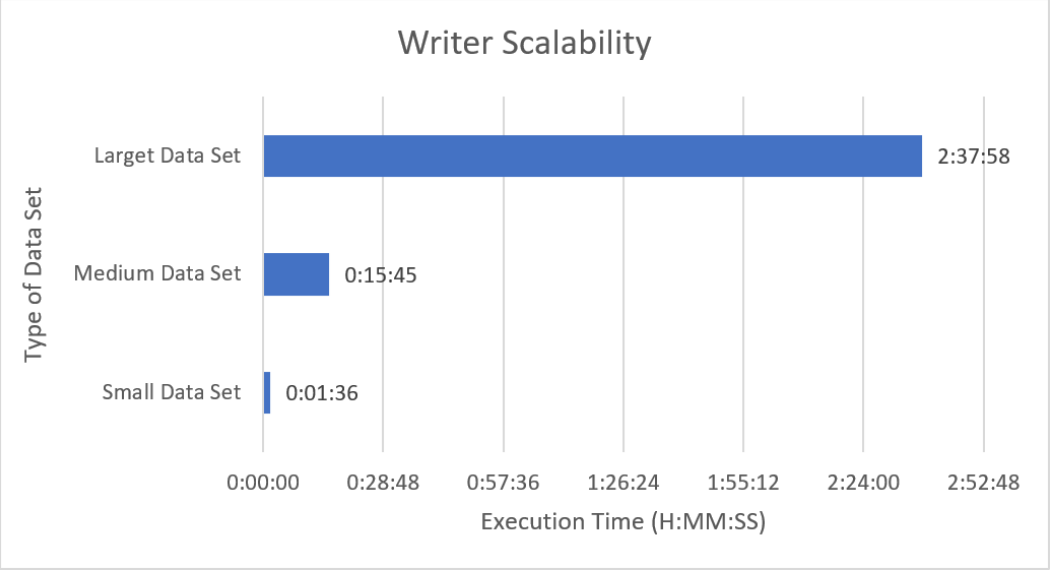
- The data set size for small, medium, and large data sets is defined based on the sizing recommendations.
- The Secure Agent is hosted on an Azure Virtual Machine in the US-EAST-2 region for all the data sets.
- The files are staged in Microsoft Azure Blob Storage.
- The number of Data Warehouse Units for Microsoft Azure Synapse SQL is set to 400c.
- The batch size is equal to the number of records in the file.
- DTM buffer size: 2147483648
- Default buffer block size: 10485760
- Heap size: Default

Environment

Virtual Machine	Azure Virtual Machine
VCPU	8 VCPUs
Memory	28 GB RAM

Performance graph

The following graph shows the effect of increasing the data volume on the execution time of writer mapping:



Conclusion

Based on the case study, linear scalability is observed for the writer mapping. When you increase the data volume, the execution time also increases linearly.

Case Studies

Refer to the following case studies for a general idea of the performance numbers.

The following configurations are considered for all the case studies:

- The data set size is 7.3 GB and 60,000,000 records.

- All mappings are passthrough.

Case studies: Gzip compression

The following case studies show the effects of gzip compression on the performance:

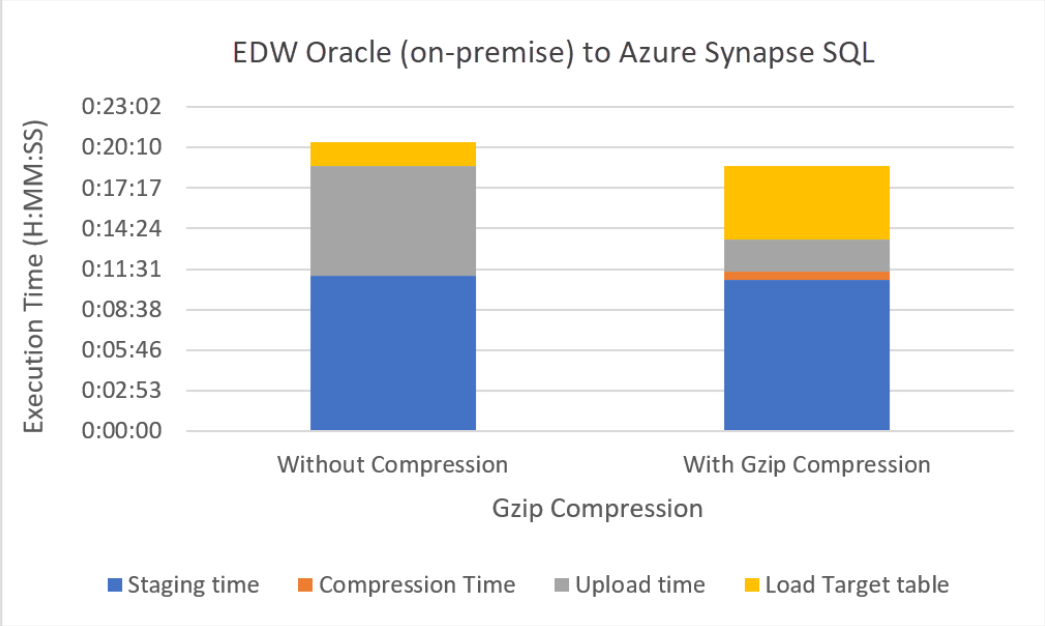
Effect of compression on data migration from on-premises Oracle Enterprise Data Warehouse to Microsoft Azure Synapse SQL

The case study shows the performance when you migrate data, with and without GZIP compression, from on-premises Oracle Enterprise Data Warehouse to Microsoft Azure Synapse SQL.
The following configurations are considered for the case study:

1. The source is Enterprise Oracle in the South India region and the target is in the US-EAST-2 region.
2. The Secure Agent is hosted on an on-premises Linux server in the same region as the source, with the capacity of 48 CPUs and 131 GB memory.
3. The files are staged in Microsoft Azure Blob Storage.
4. The number of Data Warehouse Units in Microsoft Azure Synapse SQL is set to 500c.
5. Batch size: 60000000.

Performance graph

The following graph shows the effect of gzip compression on the data migration from on-premises Oracle Enterprise Data Warehouse to Microsoft Azure Synapse SQL:



Staging time: The time taken to download data from the source table and stage the data in the Agent environment.

Compression time: The time taken to compress the staged file in the Agent environment.

Upload time: The time taken to upload the staged file to the Blob location.

Conclusion

Based on the case study, it is observed that gzip compression reduces the execution time. The execution time for the mapping without compression is `0:20:34` minutes and with gzip compression is `0:18:53` minutes.

The time to load the target table can be further improved by scaling the Data Warehouse Units in Microsoft Azure Synapse SQL to 1000c and above.

The execution time might vary in other environments. The execution time depends on the following parameters:

- The capacity of the Secure Agent hosted virtual machine in terms of CPU cores and memory.
- The throughput and IOPS of the OS disks attached to the virtual machine.
- The network bandwidth and distance between the geographical locations of source and target.

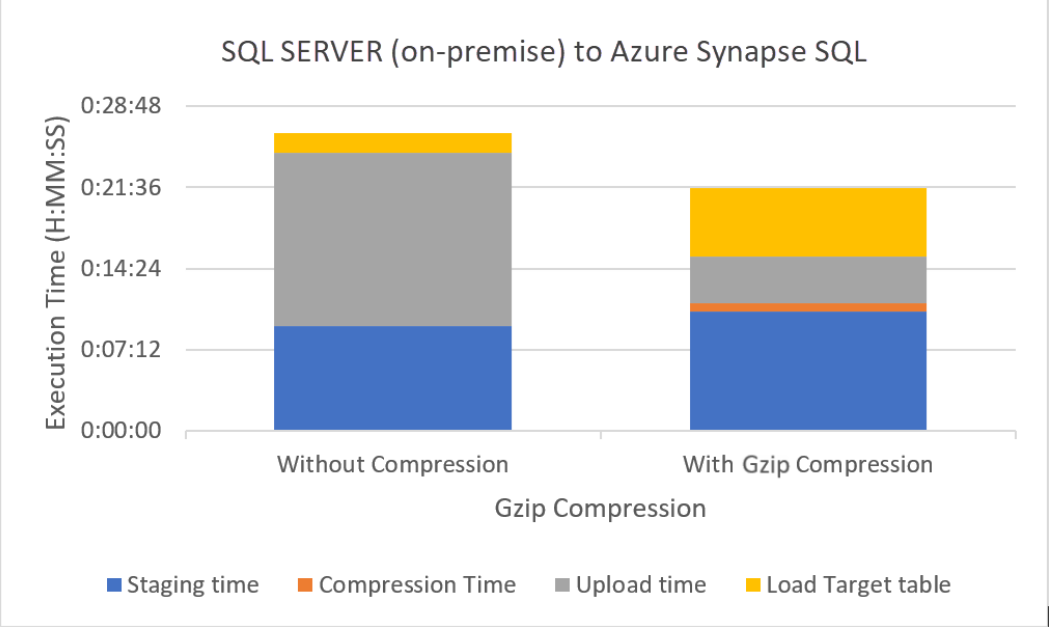
Effect of compression on data migration from on-premises Microsoft SQL Server 2017 to Microsoft Azure Synapse SQL

The case study shows the performance when you migrate data, with and without GZIP compression, from on-premises Microsoft SQL Server 2017 to Microsoft Azure Synapse SQL.

- The source is SQL SERVER in South India Region and target is Microsoft Azure Synapse SQL hosted in the US-EAST-2 region.
- The Secure Agent is hosted on an on-premises Linux server in the same region as Source, with the capacity of 48 CPUs and 131 GB memory.
- The files are staged in Microsoft Azure Blob Storage.
- The number of Data Warehouse Units in Microsoft Azure Synapse SQL is set to 500c.
- Batch size: 60000000.

Performance graph

The following graph shows the effect of gzip compression on the data migration from on-premises Microsoft SQL Server 2017 to Microsoft Azure Synapse SQL:



Conclusion

Based on the case study, it is observed that gzip compression reduces the execution time. The execution time for the mapping without compression is `0:26:25` and with gzip compression is `0:21:32`.

The time to load the target table can be further improved by scaling the number of Data Warehouse Units in Microsoft Azure Synapse SQL to 1000c and above.

The execution time might vary in other environments. The execution time also depends on the following parameters:

- The capacity of the Secure Agent hosted virtual machine in terms of CPU cores and memory.
- The throughput and IOPS of the OS disks attached to the virtual machine.
- The network bandwidth and distance between the geographical locations of source and target.

Case study: Partitioning

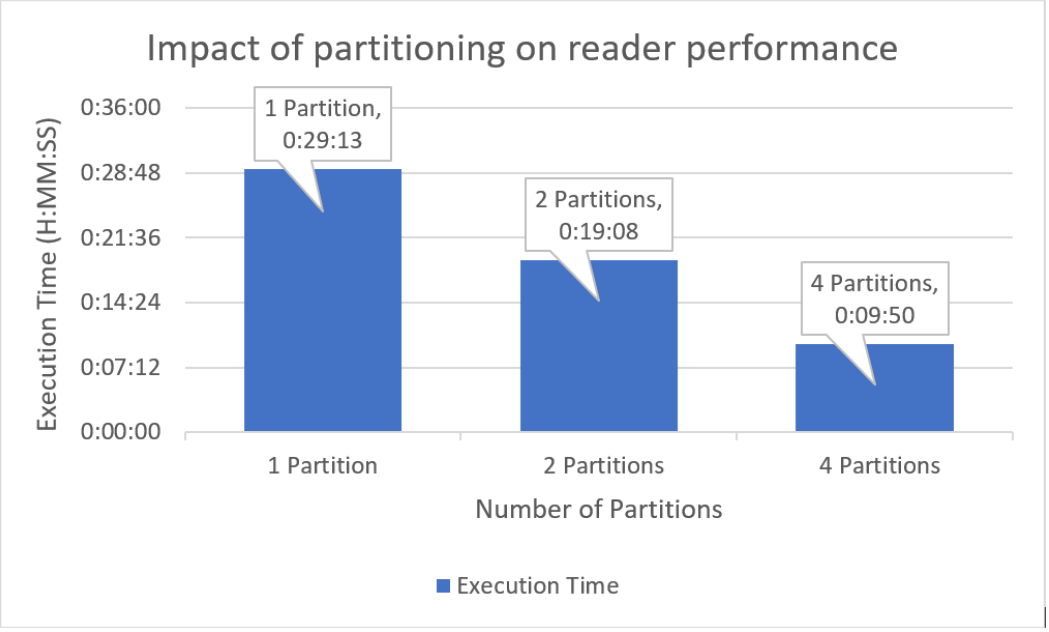
The case study shows the performance scale on increasing the number of partitions in Microsoft Azure Synapse SQL reader.

The following configurations are considered for the case study:

1. The source is Microsoft Azure Synapse SQL hosted in the US-EAST-2 region. The target is a flat file data object on an Azure Linux Virtual Machine hosted in the US-EAST-2 region.
2. The Secure Agent is hosted on the same Azure Linux Virtual Machine as the target, with the capacity of 8 VCPUs and 28 GB memory.
3. The number of Data Warehouse Units in Microsoft Azure Synapse SQL is set to 400c.
4. The files are staged in Microsoft Azure Blob Storage.
5. Compression: None.

Performance graph

The following graph shows the impact of partitioning on the reader performance:



The number of partitions are scaled up to 4 based on the source data set and configuration of the Secure Agent hosted virtual machine.

Conclusion

Based on the case study, it is observed that when you increase the number of partitions, the execution time for the reader mapping decreases.

Note: Linear scalability of partitions depends on the DWUs configured in Microsoft Azure Synapse SQL and the number of CPUs available in the Secure Agent machine. Compression is not recommended if data movement is within the network. Informatica recommends that you have the Secure Agent in the same network as source and target.

Case study: Concurrent connections to Blob Store

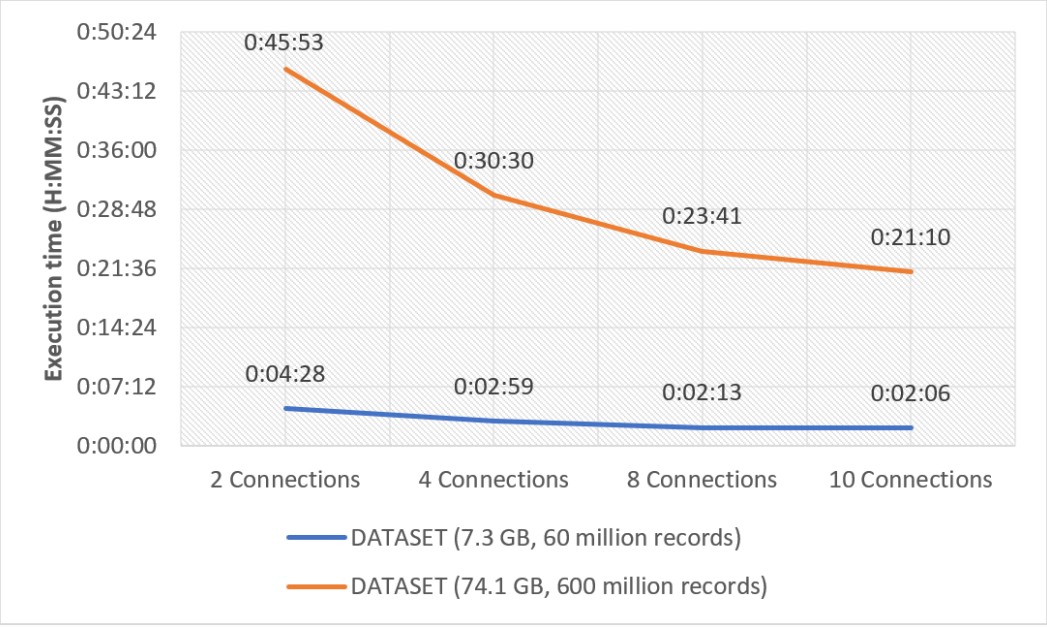
The case study shows the performance when you tune the number of Blob connections while migrating a large file from cloud virtual machine to Microsoft Azure Synapse SQL.

The following configurations are considered for the case study:

1. The source is a flat file data object on an Azure Linux Virtual Machine hosted in the US-EAST-2 region. The target is Microsoft Azure Synapse SQL hosted in the US-EAST-2 region.
2. The Secure Agent is hosted on an same Azure Linux Virtual Machine as the source, with the capacity of 8 VCPUs and 28 GB memory.
3. The number of Data Warehouse Units in Microsoft Azure Synapse SQL is set to 400c.
4. The files are staged in Microsoft Azure Blob Storage.
5. Compression: None.

Performance graph

The following graph shows the effect of tuning the number of Blob connections on the performance:



Informatica recommends that you set the number of VCPUs in the Secure Agent hosted virtual machine more than or equal to the numbers of concurrent connections to Blob Store. Mass Ingestion is recommended for data migration use cases in Azure.

Conclusion

Based on the case study, it is observed that when you increase the number of Blob connections, the execution time for the large file decreases.

A similar performance is observed when you use Microsoft Azure Data Lake Storage Gen2 to stage files.

The execution time might vary in other environments. The execution time also depends on the following parameters:

- The capacity of the Secure Agent hosted virtual machine in terms of CPU cores and memory.
- The throughput and IOPS of the OS disks attached to the virtual machine.
- The network bandwidth and distance between the geographical locations of source and target.

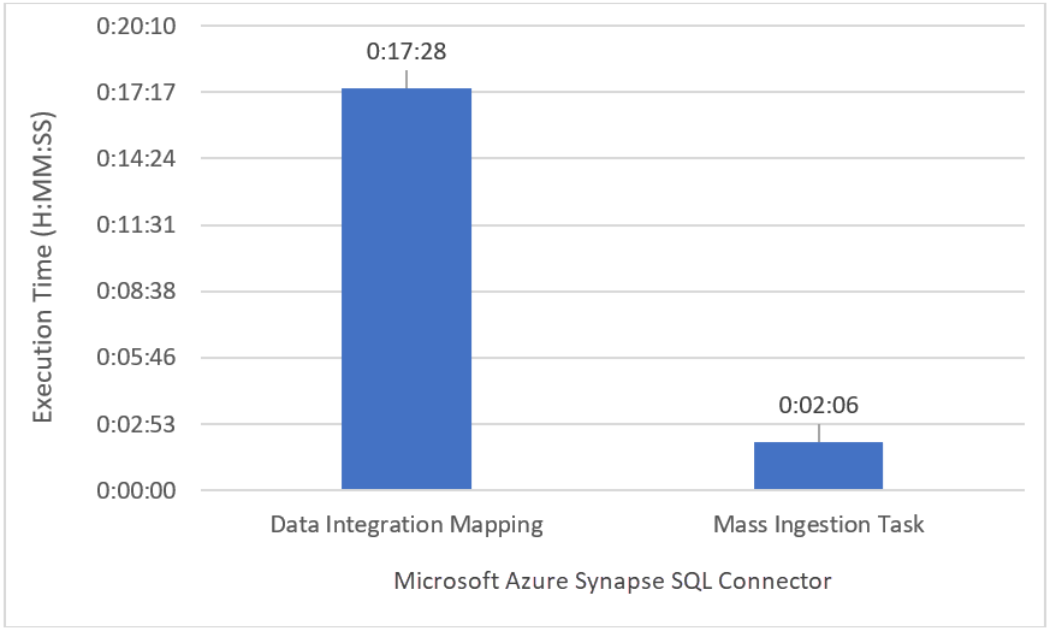
Case study: Mapping task versus mass ingestion task

The case study compares the performance of a mapping task with a mass ingestion task to transfer data.

1. The source is Microsoft Azure Blob Storage hosted in the US-EAST-2 region. The target is Microsoft Azure Synapse SQL hosted in the US-EAST-2 region.
2. The Secure Agent is hosted on the same Azure Linux Virtual Machine and same region as the source, with the capacity of 8 VCPUs and 28 GB Memory.
3. The number of Data Warehouse Units in Microsoft Azure Synapse SQL is set to 400c.
4. The files are staged in Microsoft Azure Blob Storage.
5. Number of concurrent connections to Blob Store: 4
6. Batch size: 60000000
7. Compression: None.

Performance graph

The following graph compares the performance of a mapping task with a mass ingestion task:



Conclusion

Based on the case study, a significant decrease in the execution time is observed when you use a mass ingestion task to transfer files from a Azure Blob endpoint to Microsoft Azure Synapse SQL when no data processing is required, as it saves the overhead of local and Microsoft Azure Blob staging.

A similar performance is observed when you use Microsoft Azure Data Lake Storage Gen2 to stage files.

The execution time might vary in other environments. The execution time also depends on the following parameters:

- The capacity of the Secure Agent hosted virtual machine in terms of CPU cores and memory.
- The throughput and IOPS of the OS disks attached to the virtual machine.
- The network bandwidth and distance between the geographical locations of source and target.

Case studies: Pushdown optimization

To obtain optimal performance, the mapping design plays a key role in the pushdown performance.

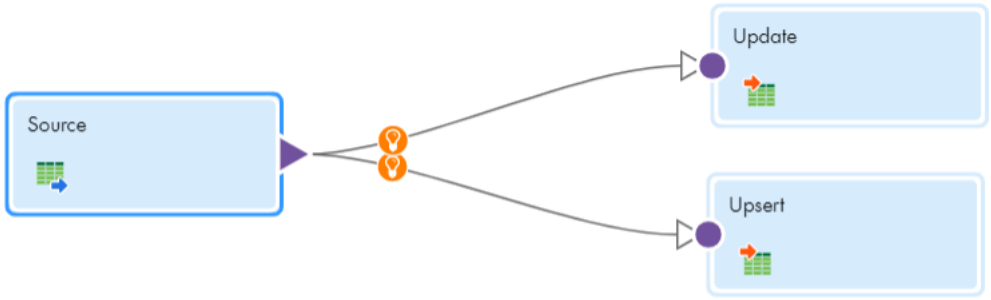
For example, to achieve optimal performance when you enable source pushdown optimization for a mapping that involves multiple transformations, you must design the mapping in such a way that heavy transformations, such as the Joiner transformation, must be placed further downstream closer to the Target transformation.

The following case studies show the results of mappings configured with full pushdown, source pushdown, and partial pushdown optimization:

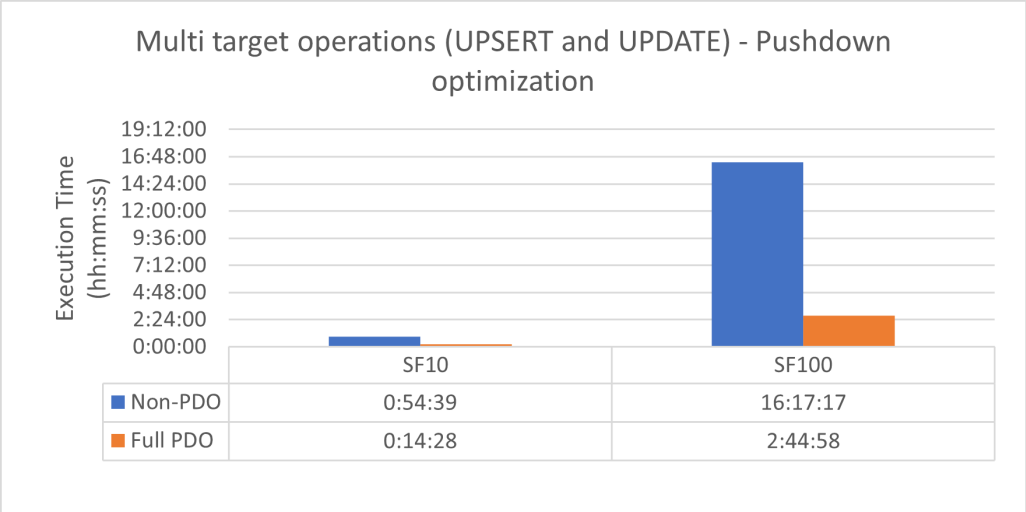
Case 1

In this scenario, the mapping consists of a single Microsoft Azure Synapse SQL table as the source and multiple Microsoft Azure Synapse SQL targets configured with different operations such as insert, delete, upsert, and update. The mapping runs with full pushdown optimization.

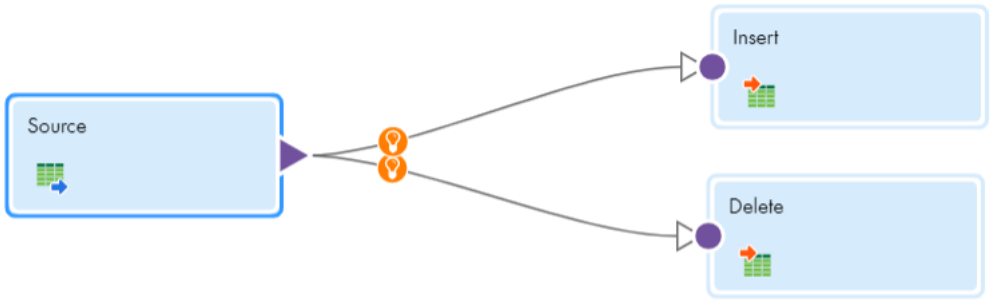
The following figure shows the mapping configured with update and upsert operations and enabled for full pushdown optimization:



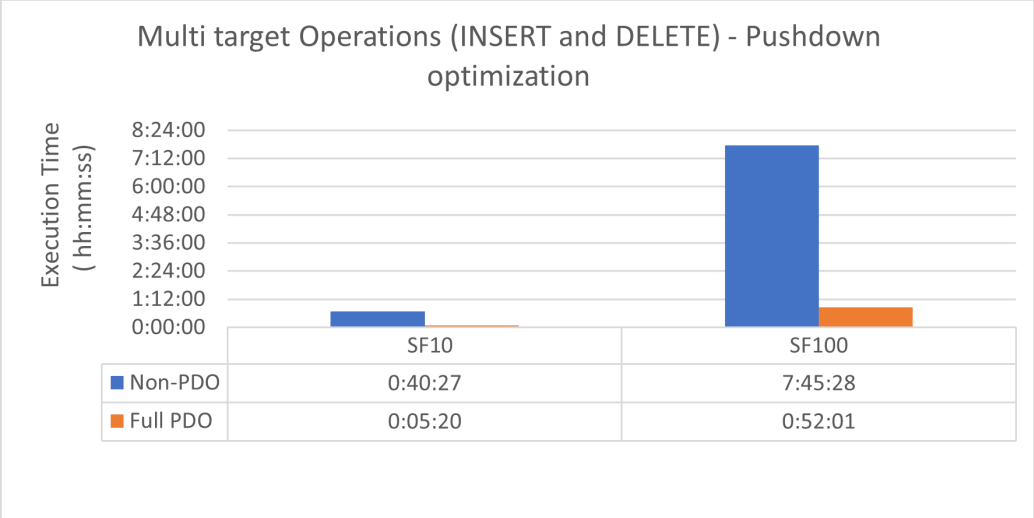
The following figure shows the results when you run the mapping configured with update and upsert operations and enabled with and without pushdown optimization:



The following figure shows the mapping configured with insert and delete operations and enabled for full pushdown optimization:



The following figure shows the results when you run the mapping configured with insert and delete operations and enabled with and without pushdown optimization:

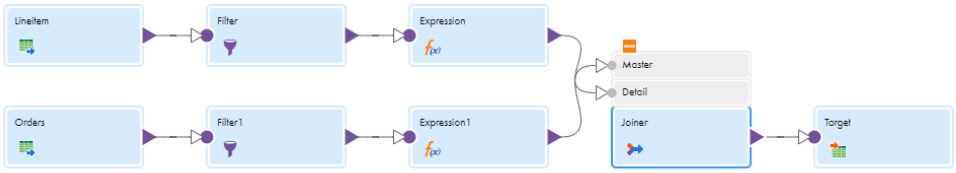


The results show that full pushdown optimization with multitarget database operations show a significant performance gain when compared to the same mapping without pushdown optimization.

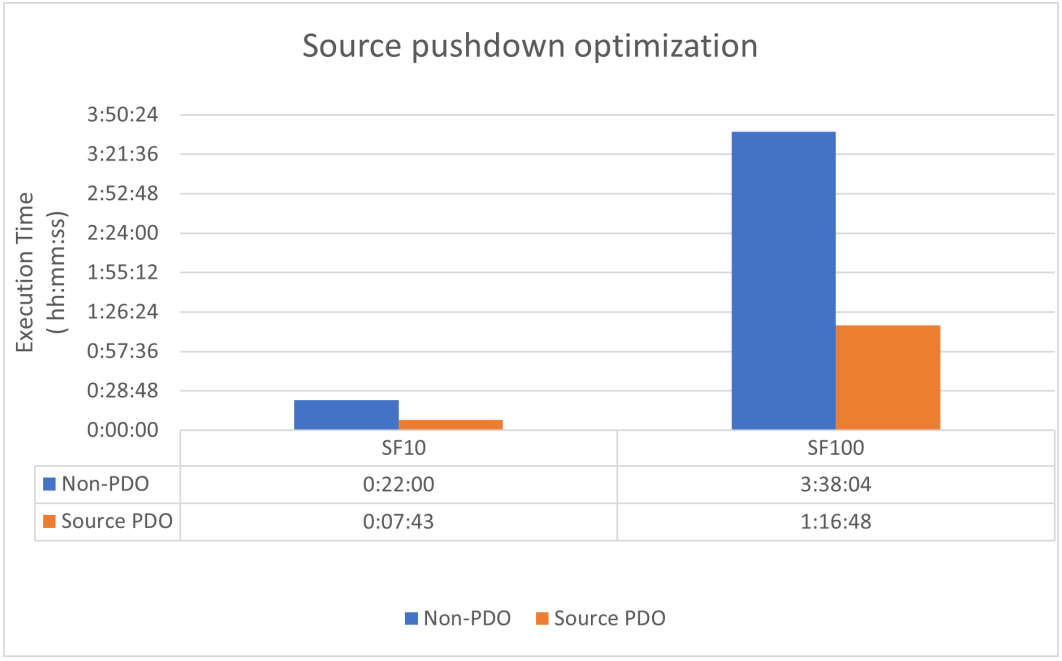
Case 2

In this scenario, the mapping includes multiple Microsoft Azure Synapse SQL sources. The mapping also includes a series of parallel transformations such as simple expressions and filters, followed by a Joiner transformation. The mapping runs with source pushdown optimization.

The following figure shows the mapping configured with source pushdown optimization:



The following figure shows the results when you run the mapping with and without pushdown optimization:

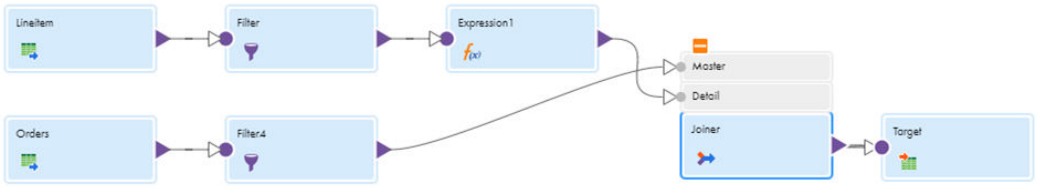


The results show that the mapping performance with source pushdown optimization is approximately 3x times faster when compared to the same mapping without pushdown optimization.

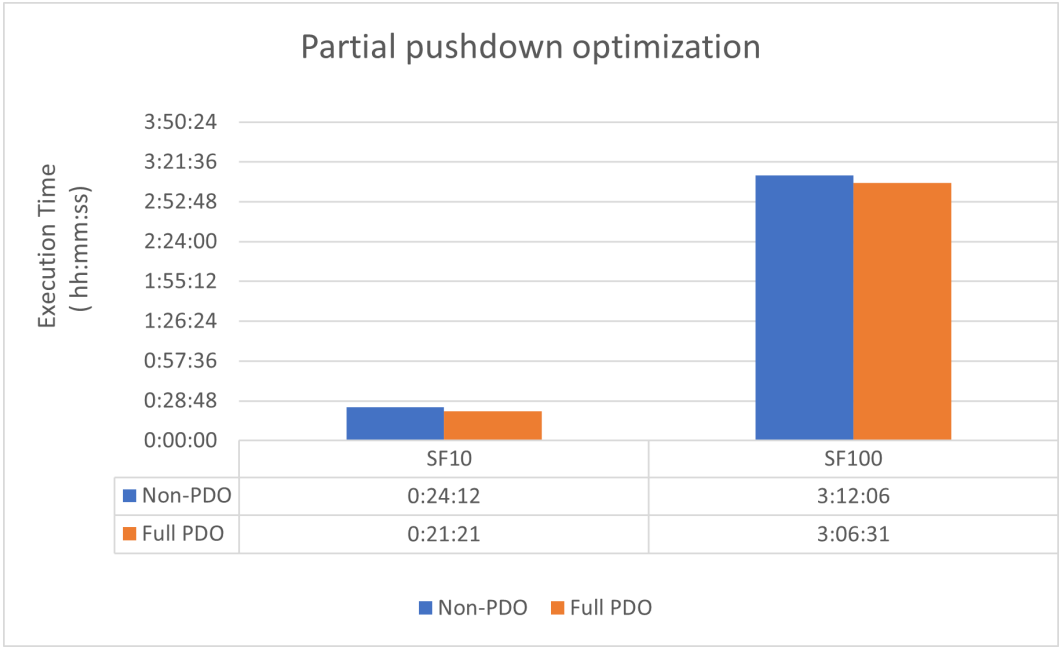
Case 3

In this scenario, the mapping configured with full pushdown optimization runs with partial pushdown optimization.

The following figure shows the configured mapping with full pushdown optimization:



The following figure shows the results when you run the mapping with and without pushdown optimization:



The results show that there is not much performance gain in this scenario when the mapping runs with full pushdown optimization as compared to a mapping without pushdown optimization.

Performance benefits with pushdown optimization

When you use Microsoft Azure Synapse SQL Connector, you can observe the following performance benefits while using source or full pushdown optimization in comparison to a mapping without pushdown optimization:

- Full pushdown optimization provides better performance for all supported transformations and operations.
- Source pushdown optimization comparatively does not always provide better performance for all supported transformations and their combinations when compared to a mapping that is not enabled for pushdown optimization. To obtain optimal performance with source pushdown, the mapping design plays a key role.
- Partial pushdown shows negligible performance benefits.
- Scaling the Microsoft Azure Synapse SQL data warehouse provides better performance.

Authors

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